RENSSELAER MECHATRONICS

DC Motor Parameter Identification

Part 1: Motor Steady state response

Objectives:

- Perform different experiments on a DC motor to determine its motor parameters

Background Information:

A DC motor can be modeled with an electrical system coupled to a mechanical system by a magnetic field. The equation for the electrical system is:

$$V = L\frac{di}{dt} + Ri + K_b\omega$$

Where:

- K_b – back emf constant
- R – armature resistance
- L – inductance
- i – the current through the motor windings

The coupling is seen from the voltage generated by the spinning motor $K_b\omega$. The equation for the electrical system is

$$K_t i = J\frac{d\omega}{dt} + \omega b$$

- K_t – torque constant
- b – viscous damping coefficient
- J – armature Inductance (or combined motor and load Inertia if load is attached)
- ω – the velocity of the motor shaft

In the steady-state (no change with time) the equations become:

$$V_{ss} = Ri_{ss} + K_b\omega_{ss}$$

$$K_t i_{ss} = b\omega_{ss}$$

These equations, with the steady state current and voltage measurements can be used to determine the motor parameters.

Parameter Estimation Background

Resistance: If there is access to the motor leads this can be measured with a multimeter. This is not the most reliable method. Typically, this is done measuring no-speed voltage and current to estimate the resistance.

Torque and Back EMF constants:

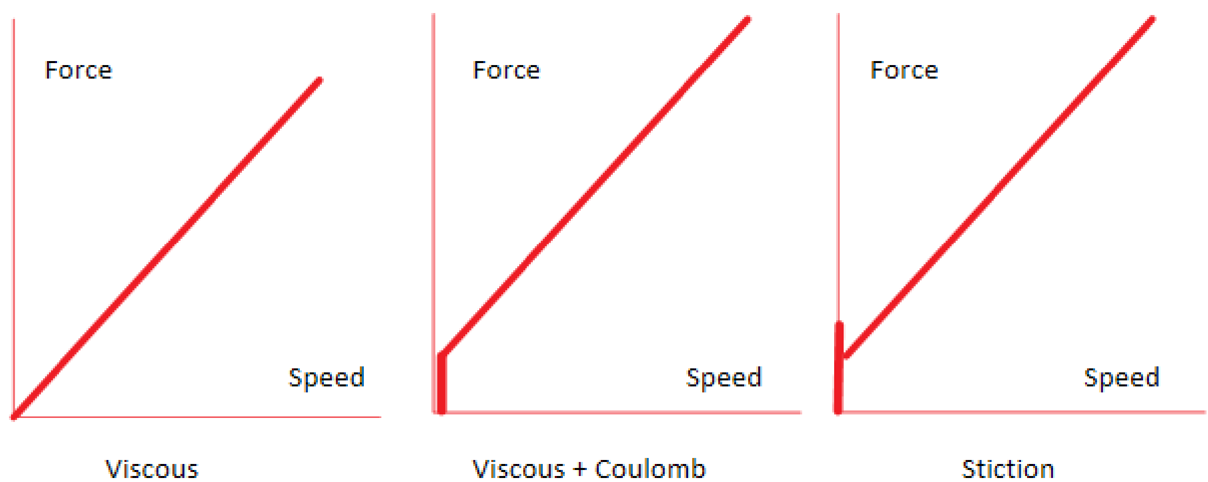
First assume that energy is conserved in the motor in which case $K_b = K_t$ and only one needs to be estimated. From the steady-state electrical equation:

$$V_{ss} = Ri_{ss} + K_b\omega_{ss}$$

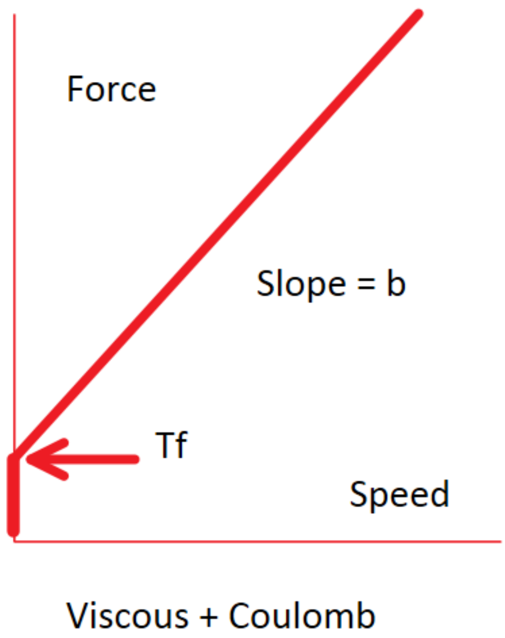
If resistance is known, speed, current, and voltage are measured these equations can be used to calculate K_b which, by assumption equal to K_t. For *gearmotors*, since they contain a gear train and have significant losses K_t must be estimated. For a LEGO NXT motor is about 65% of K_b. This value can be obtained by an experiment where a known load is applied, and the current is measured.

Viscous and Coulomb Friction Estimation:

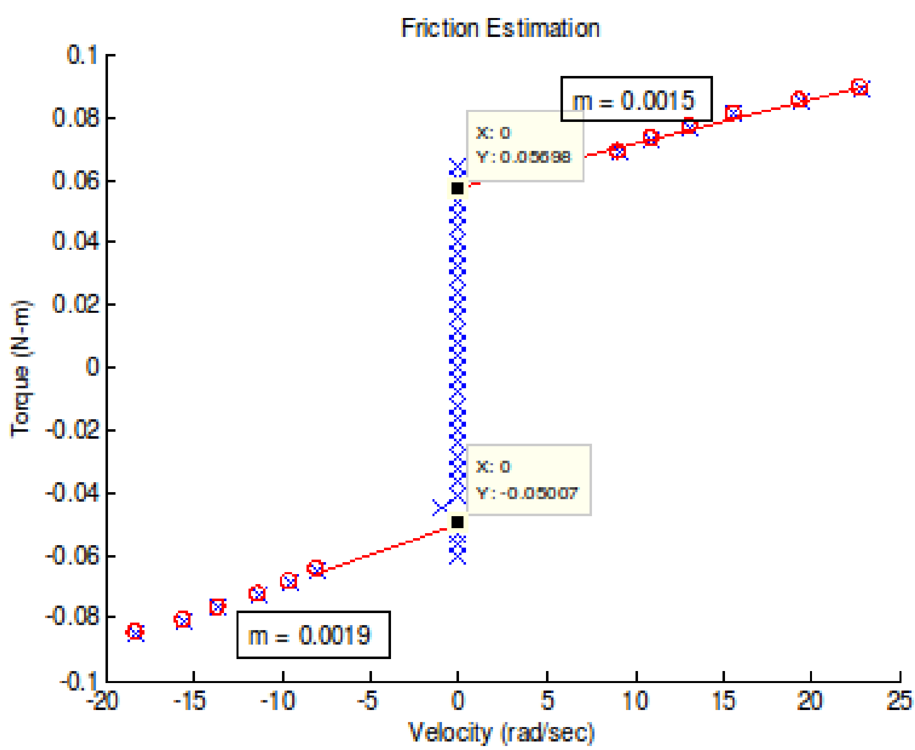
There are typically 3 different components of friction: viscous, coulomb, and static. This leads to three different types of friction models:



For a motor these curves can be obtained by plotting steady state speed and measured torque $K_t i$.



An experimental example is:

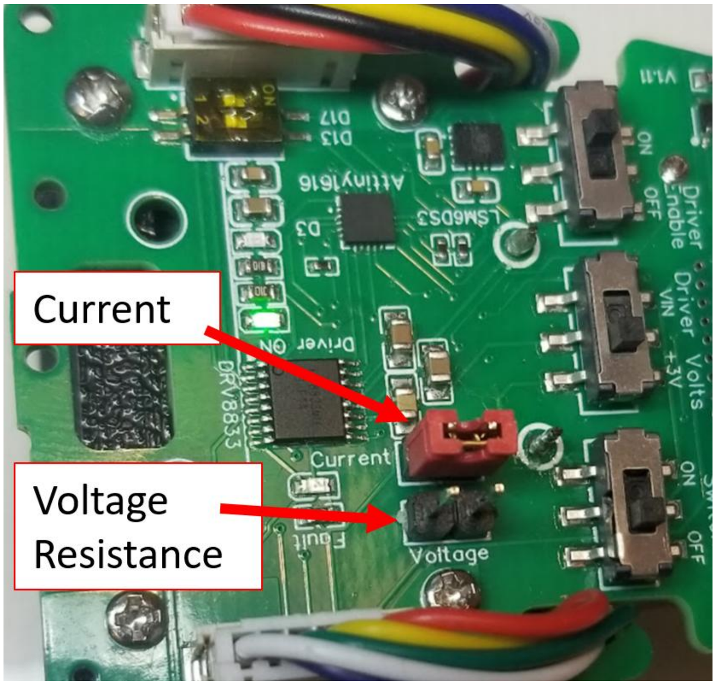


This will require steady state measurements of voltage and current at a couple different speeds. Once this graph is obtained then the viscous damping coefficient b is the slope of the line, and the constant coulomb friction is the y-intercept.

System Measurements

Resistance and Voltage:

Resistance and voltage can be measured by using a multimeter and measuring between the two motor leads which are conveniently routed to the two pins above the "Voltage" label on the board.



The red jumper should ONLY be placed on the Current pins. It should NEVER be placed on the Voltage pins – this will short out the motor pins and could damage the multimeter or the board.

To Measure voltage:

- Place the multimeter in the 20V setting
- Verify the multimeter is set up to measure voltage. If it set up to measure current and you connect it to the voltage pins you will short out the motor pins and it could damage the multimeter or the board.
- Place the multimeter leads on the Voltage pins (the current jumper must be on)

The figure below shows a multimeter connected the voltage pins via jumpers to measure voltage:

If you want to measure resistance temporarily remove the connection to the Voltage pins, select the 200 ohm setting on the multimeter, and reconnect to the voltage pins.

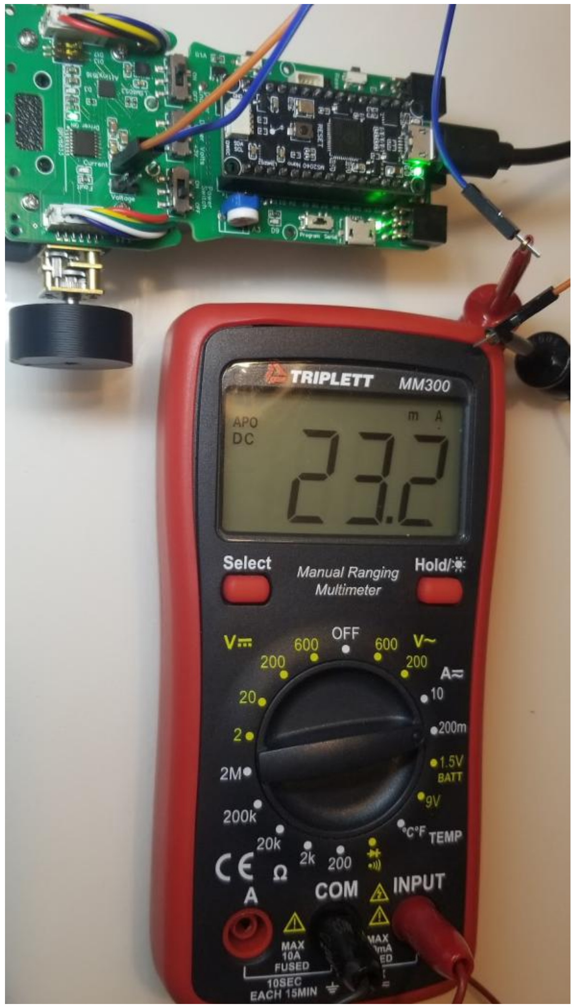
<u>Current:</u>

To measure current one of the motor wires must be cut, and a multimeter must be placed between these wires. The Jumper on the pins labeled Current allows an easy way to cut one of the wires going to the motor so a multimeter can be placed between.

To measure current:

- Place the multimeter in the 200m current setting (ONLY use this setting if you KNOW the current will be less than 0.2A)
- Remove the jumper on the Current pins (current will stop flowing to the motor and it will stop spinning)
- Place the multimeter on the Current pins (current will flow again and the motor will resume spinning)

The figure below shows the setup to measure current less than 0.2A:



Typically, a rough value for resistance can be made by measuring the resistance across the motor leads. Then use this estimate of R to estimate the current you will expect for an applied voltage. When in doubt start off at a small voltage (current) to verify.

If you will be above 200ma (0.2A) use the 10A setting on the multimeter shown below:

DC Motor measurements – Estimate R:

Make sure the system is not connected to the computer. Use the multimeter to directly measure the resistance of the motor. Set up the multimeter to measure 200 ohms and connect the multimeter to the voltage pins. Make a measurement, then rotate the motor shaft and make another one. You may notice that the multimeter will not display any resistance for a couple seconds – this is because you created a voltage when you turned the motor. Move the motor several times making several measurements, record your average resistance below

R_avg (direct measurement, several times) = ___________

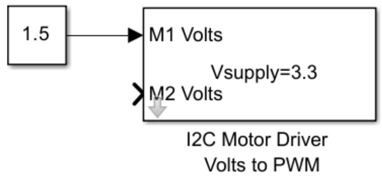
Calculate the expected current with your R_avg if you apply 1 volts and 2 volts to the motor (I=V/R):

Current Expected for 1 volt = _________________

Current Expected for 2 volts = _________________

Next, estimate resistance by applying a voltage, holding the motor shaft, and measuring voltage and current.

- Set up the multimeter to measure voltage, connect it to the voltage pins
- Run the following Simulink diagram to apply 1.5 specified volts to the motor:



- Once the motor starts spinning hold the system and stop the wheel from spinning. Measure the voltage.

  Repeat this procedure for a specified voltage of 2.0.

  Measured Voltage (1.0specified volts, wheel not spinning): ________________
  Measured Voltage (2.0 specified volts, wheel not spinning): ________________

Repeat the same two specified voltages, but now measure the current at each voltage when you are holding the wheel

- Remove the multimeter from the voltage pins
- Set up the multimeter to read current
- Remove the current jumper
- Attach the multimeter to the current pins
- Run the Simulink diagram with 1.0 specified voltage.
- Once the motor starts spinning hold the system and stop the wheel from spinning. Measure the current.

  Repeat this procedure for a specified voltage of 2.0.

  Measured Current (1.0 specified volts, wheel not spinning): ________________
  Measured Current (2.0 specified volts, wheel not spinning): ________________

Calculate R for 1 volt: ________________
Calculate R for 2 volts: ________________
Average R: ________________

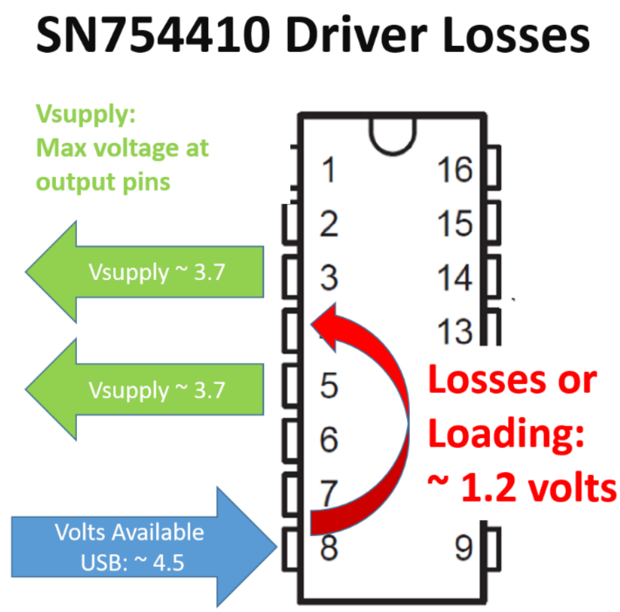
How does your Resistance compare to the direct measurement?

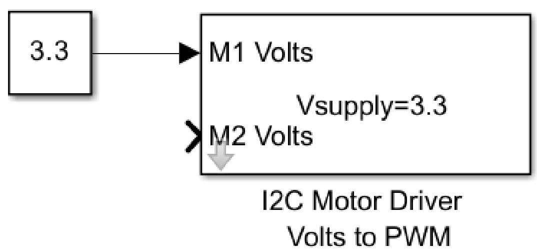
Driver Supply Voltage (direct measurement):

The Motor driver block computes the approapriate PWM value based on the desire voltage. To do this it has to scale the actual available voltage at the ouput of the actual driver, to the correct PWM value.

The value of Vsupply in this block needs to be the maximum voltage available to your device, in this case the motor. To find this value turn the driver completely ON, and measure the voltabe at the output pins of the driver. This value is the maximum voltage you can obtain from the output of the driver Vsupply.

To find this, run the following Simulink diagram:





Since the Vsupply is 3.3 and the input is 3.3 the PWM duty cycle will be 100% (fully on).

- Make sure the “Driver volts switch” is set to 3V
- With the motor connected and freely spinning measure the voltage at the voltage pins

Vsupply (Driver volts switch set to 3V) = _______________

Next see how much voltage is available when USB power us used:

- Make sure the “Driver volts switch” is set to VIN (USB voltage)
- With the motor connected and freely spinning measure the voltage at the voltage pins

Vsupply (Driver volts switch set to VIN) =_______________

- Make sure the “Driver volts switch” is set to back to 3V.

These will be the Vsupply values when using USB voltage or 3.3V of the system.

When you add batteries to the system, you will need to repeat this process to determine the battery voltage you have available to the system.

Speed, Voltage and current:

The current and voltage need to be measured at different speeds to calculate the back EMF constant K_b. The Simulink diagram below can be used in external mode at 0.03 seconds to record the steady state velocity. This same diagram can be used later to capture the step response.

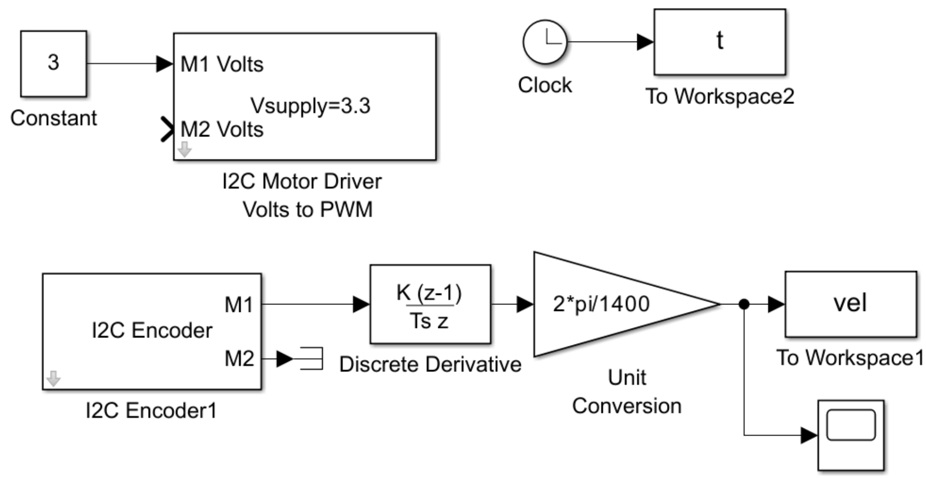


Figure 1: Simulink diagram for speed measurements

Data Collection:

- Run the motor at least 3 different steady state speeds, and record the speed data (1, 2, and 3 volts would be recommended)
  - At the steady state speeds measure the velocity and current. This may require multiple tests. Record voltage and current at each speed.

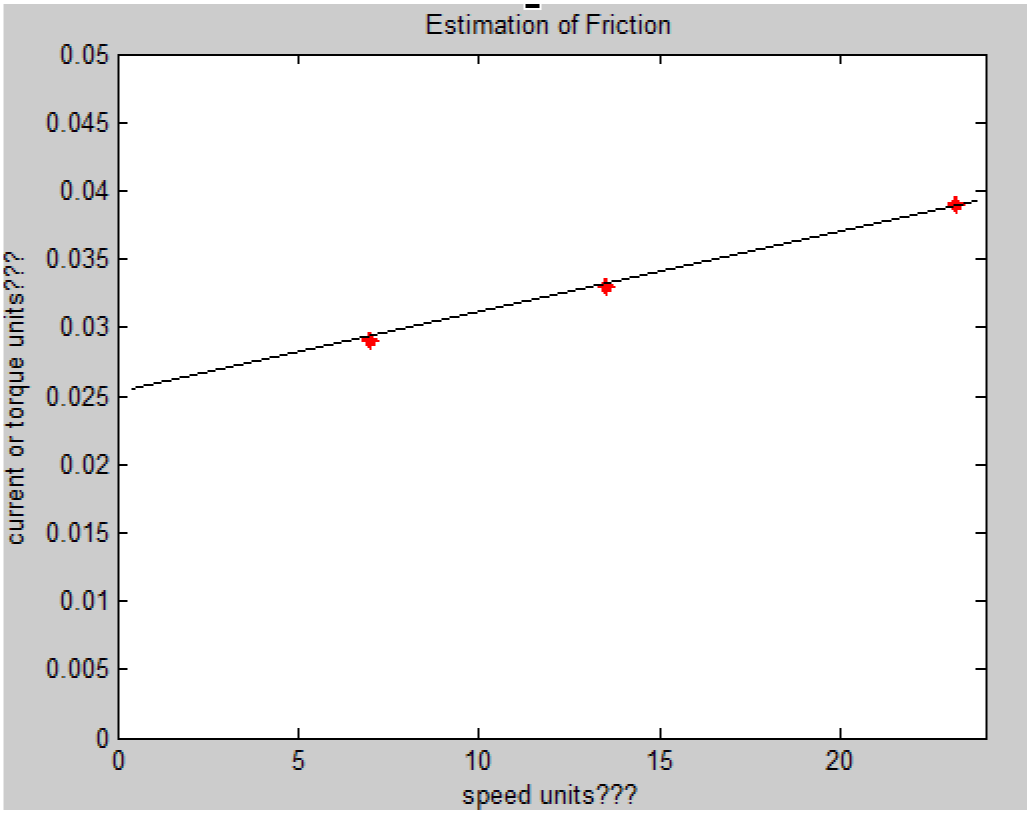
$V_{1specified}$ = ________ , $V_{1meaured}$ =_______, ω_1 =__________, I_1 =__________
$V_{2specified}$ = ________ , $V_{2meaured}$ =_______, ω_2 =__________, I_2 =__________
$V_{3specified}$ = ________ , $V_{3meaured}$ =_______, ω_3 =__________, I_3 =__________

Parameter Identification

- Use the collected data and the steady state equation to determine K_b from measured speed, current and voltage at the 3 different speeds:
  - $V_{ss} = Ri_{ss} + K_b\omega_{ss}$

K_b =______________________ (specify units)

- Use the data to plot the speed vs. torque curve (with the correct units) to determine the coulomb friction value, and the damping coefficient. Assume K_t is about 70% K_b. An example graph is shown below (units and data are not correct):



Questions:

- Create a plot of current (amps, y-axis) versus speed (rad/sec. x-axis). Clearly label the axes and provide a title for the graph
- Do the measured voltages match the voltage set in the Simulink diagram when you use the multimeter to measure the voltage?
- What are your experimental values for (with units!!, use R measured with voltage and current):
  - R, K_b, b, T_f, K_t?

Part 2: Motor Transient Response

From the mechanical and electrical DC motor equations the 2nd order transfer function for a DC motor is:

$$G(s) = \frac{\omega(s)}{V(s)} = \frac{K_t}{LJs^2 + (JR + Lb)s + (Rb + K_tK_b)}$$

If the inductance is "small" it can be neglected (L=0) and the first order transfer function is obtained:

$$G(s) = \frac{\omega(s)}{V(s)} = \frac{K_t}{JRs + (Rb + K_tK_b)}$$

Where the motor parameters are:

- K_t – torque constant
- K_b – back emf constant

- b – viscous damping coefficient
- R – armature resistance
- J – armature inductance (or combined motor and load inertia if load is attached)
- L – inductance

The equation in time constant form is:

$$G(s) = \frac{\omega(s)}{V(s)} = \frac{K_t/(Rb + K_tK_b)}{JR/(Rb + K_tK_b)s + 1} = \frac{K}{\tau s + 1}$$

- **Time Constant**: $\tau = \frac{JR}{Rb+K_tK_b}$
- **Steady State Gain**: $K = \frac{K_t}{Rb+K_tK_b}$

From this equation the time constant and steady state gain can be identified. Calculate K from the parameters you identified (include units):

- **Steady State Gain**: $K = \frac{K_t}{Rb+K_tK_b}$ = ____________________ (Units _______)

Questions:

- What is your calculated values for the steady state gain K?

Parameter Estimation from step response

Obtain the step response of the motor. An example step response is shown below (you may already have this data from a previous lab). (earlier versions of MATLAB would need to obtain this data with serial mode, later versions (2020a) only use Monitor and Tune (external) mode).

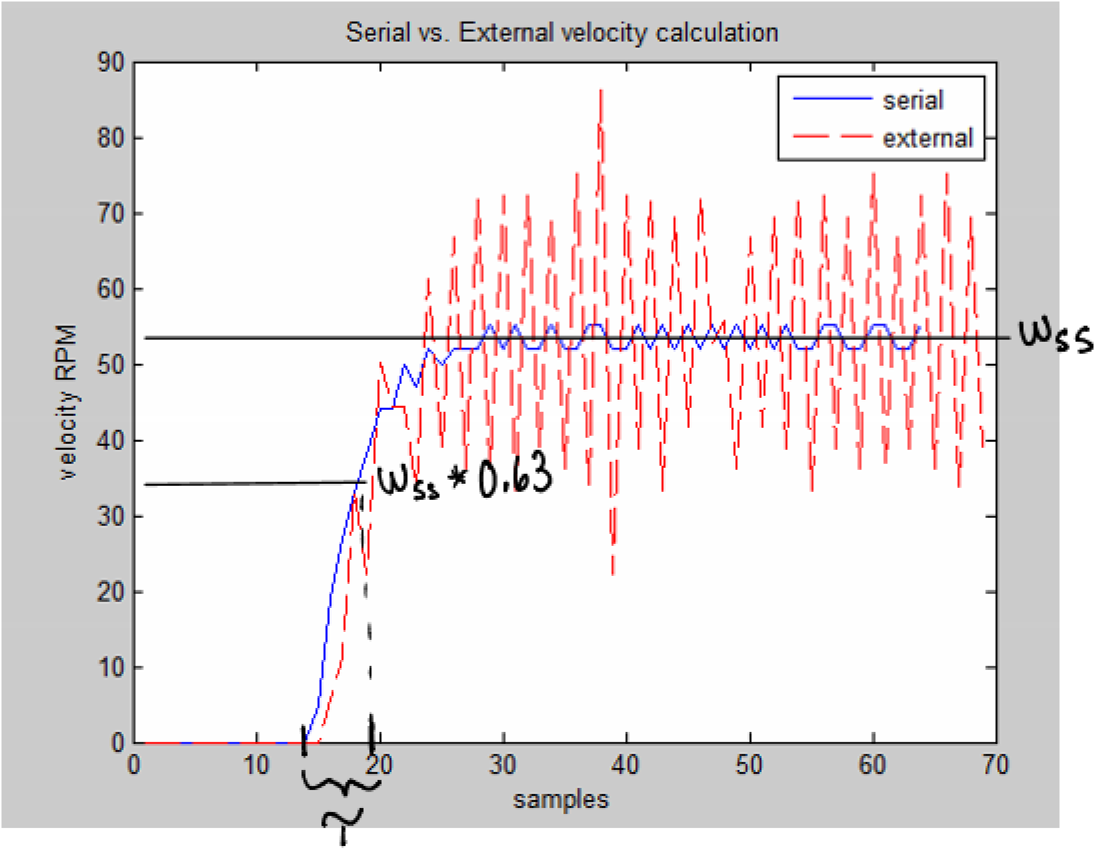


Figure 2: Sample Step response data obtained with 0.03 second sample time, 3 volt step

Take note of what voltage you used to generate this step response V_{in} (this should be the actual measured voltage):

V_{in} = ________________

Calculate the experimental steady state gain (the step must be from zero speed as in the graph above):

$K_{exp} = \omega_{ss}/V_{in} =$___________________

Where ω_{ss} is the steady-steady state value from the step response graph, and V_{in} is the voltage used to generate the step response.

Next experimentally compute τ by finding how long it takes for the step response to reach 63% of its steady state value:

$\tau_{exp} =$___________________

The only unknown in the formula for the time constant τ is the inertia. Use the formula for the step response to calculate the inertia J:

$J =$______________________ (be sure to include units)

Questions:

- What is your experimental values for the time constant and steady state gain τ, K?
- What is the inertia identified from the time constant?
- What is the percent error from the calculated K versus the K_{exp} value from the step response?
- Kt can be determined by applying known loads and measuring the current. For this motor it is determined that Kt is approximately 65% of Kb. Why?
- If all the linear motor parameters are determined from separate experiments, the computed time constant for the resulting linear model is approximately $\tau = .042$. How does your value for the time constant compare to this - why?